

Assignment Set: K-Means Clustering

Assignment 1: Basic Concepts of K-Means

Question 1 Explain the following terms related to K-Means clustering:

1. Clustering
2. Centroid
3. Euclidean Distance
4. Iteration
5. Convergence

Question 2

Differentiate between:

- Supervised learning and unsupervised learning
- Classification and clustering

Question 3

Write the advantages and disadvantages of K-Means clustering.

Assignment 2:

Consider the following 2D data points:

Point	X	Y
A	1	1
B	2	1
C	4	3
D	5	4

Assume:

- Number of clusters $K = 2$
 - Initial centroids:
 - $C_1 = (1,1)$
 - $C_2 = (5,4)$
- a) Calculate the Euclidean distance of each point from both centroids.
b) Assign each point to the nearest cluster.

- c) Compute the new centroids after the first iteration.
- d) Check whether the algorithm has converged.
- e) Explain Elbow method to compute the optimal value of K in k-means algorithm.

Assignment Set: K-Nearest Neighbors (KNN)

Assignment 1:

Question 1

Define the following terms:

1. K-Nearest Neighbors (KNN)
2. Instance-based learning
3. Distance metric
4. Classification
5. Regression

Question 2

Differentiate between:

- KNN Classification and KNN Regression
- Supervised learning and Unsupervised learning

Question 3

Write the advantages and disadvantages of the KNN algorithm.

Assignment 2: Distance Measurement

Consider the following data points:

Point	X	Y	Class
A	1	1	Red
B	2	2	Red
C	4	4	Blue
D	5	5	Blue

A new point is given:

$$P = (3,3)$$

Tasks

Question 1

Calculate the Euclidean distance between P and each training point using:

Question 2

Find the 3 nearest neighbors of point P .

Question 3

Predict the class of point P using majority voting.

Assignment 3:

Question 1

Write the step-by-step KNN classification algorithm.

Question 2

Explain the role of the parameter K in KNN.

Question 3

What happens if:

1. $K = 1$
2. K is very large
3. The dataset contains noise